Data-free physics-informed neural networks for the linear Einstein–Boltzmann equations of the cosmic microwave background

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[Draft abstract; final numbers pending.] We present a physics-informed neural network (PINN) that solves the linearised Einstein–Boltzmann equations of the photon–baryon–dark-matter–neutrino system in conformal Newtonian gauge without any training data: the networks are trained only on the residuals of the equations. We identify the ingredients without which training fails or converges to wrong solutions, each isolated in controlled experiments: initial conditions that satisfy the Einstein 00 constraint, the 00 residual as a loss term (which removes a constraint-violating solution that otherwise grows like conformal time), time-dependent physical scaling of the outputs, preconditioning of the stiff Thomson-scattering equations with a slow-manifold hand-over at the tight-coupling switch, and time-marching over a chain of networks (a single network collapses to a trivial solution once modes oscillate). With these, single modes reach 10^{-4}–10^{-3} accuracy against a stiff ODE solver through several acoustic oscillations, and a bundle over wavenumber reaches an RMS error of $\sim 10^{-3}$ in the tight-coupling era. A bundle that also conditions on (ω_b, ω_c) reaches RMS errors of $(0.5\text{–}1.3) \times 10^{-2}$, with the worst errors at the edges of the parameter box; there, the Einstein-00 residual flags them without reference data. [**Pending:** accuracy through recombination, C_ℓ^{TT} vs CLASS, and the beyond-ΛCDM inverse problem.]

I. INTRODUCTION

Predictions for cosmic microwave background (CMB) anisotropies rest on the linearised Einstein–Boltzmann equations, solved for every Fourier mode by codes such as CAMB [1] and CLASS [2, 3]. Two developments have changed how these predictions are used. Neural emulators [4–7] replace a Boltzmann code by a network trained on its outputs, and differentiable Boltzmann codes [8–10] provide exact gradients of observables with respect to parameters. Emulators are fast but need a solver to produce training data, which ties them to models that some code already implements. Differentiable solvers are exact but solve one cosmology at a time.

Physics-informed neural networks (PINNs) [11, 12] sit between the two. A PINN is trained on the residual of the governing equations rather than on solutions, so it needs no training data, and a single network can represent a whole family of solutions when the parameters are network inputs (a "bundle" [13]). In cosmology PINN bundles have so far been applied to the background expansion [14, 15] and to the sub-horizon growth equation for matter, a single second-order ODE [16]. The latter work describes itself as a first step towards the coupled Boltzmann system. Related neural approaches solve the homogeneous Boltzmann equations of particle decoupling [17]. To our knowledge no PINN has been applied to the coupled photon–baryon–dark-matter–neutrino perturbation system that determines the CMB.

This system is a hard test for PINNs. It is stiff (Thomson scattering is 10^3–10^9 times faster than the expansion before recombination), multi-scale (the integration covers seven decades in conformal time and the variables span many orders of magnitude), oscillatory (acoustic waves and free streaming), and it carries a constraint (the Einstein 00 equation) that the evolution equations propagate but do not enforce. This paper asks whether a data-free PINN can nevertheless reach the accuracy needed for CMB predictions, and documents what it takes.

Our contributions are:

1. A data-free PINN for the full linear system in conformal Newtonian gauge: tight-coupled photon–baryon fluid, the photon temperature and polarisation hierarchies with Thomson scattering, massless neutrinos, cold dark matter, and the metric. The network is trained as a bundle over wavenumber, and over (ω_b, ω_c) in the tight-coupling era.

2. A set of ingredients without which the training fails, each isolated in controlled experiments (Sec. IV A): constraint-consistent initial conditions, the Einstein 00 residual as a loss term, time-dependent physical scaling of the outputs, Thomson-row preconditioning with a slow-manifold hand-over at the tight-coupling switch, and time-marching over a chain of networks.

3. Line-of-sight CMB temperature spectra C_ℓ^{TT} computed from PINN source functions, validated against CLASS.

4. A demonstration outside ΛCDM: a bundle over a late-time modification of the effective Newton constant for perturbations, used as a data-free, differentiable emulator in an inverse problem.

We report all accuracy against an independent stiff ODE solver that integrates exactly the same equations, so that PINN error and modelling error are separated, and against CLASS. We also report where the method falls short of its targets and what it costs.

II. EQUATIONS AND REFERENCE SOLUTIONS

A. Perturbation equations

We work in conformal Newtonian gauge, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\mathbf{x}^2]$, with the variables and conventions of Ma & Bertschinger [18] (MB95): density contrasts δ_i, velocity divergences θ_i, anisotropic stresses σ_i, and Legendre moments $F_{\gamma\ell}$, $G_{\gamma\ell}$, $F_{\nu\ell}$ of the photon intensity, photon polarisation and massless-neutrino distributions, truncated at $\ell_{\max}$ with the MB95 scheme. The metric is evolved with the $0i$ Einstein equation,

$$\dot\phi = -\mathcal{H}\psi + \frac{4\pi G a^2}{k^2}\sum_i (\bar\rho_i + \bar P_i)\theta_i, \tag{1}$$

and ψ follows algebraically from the traceless ij equation, $k^2(\phi-\psi) = 12\pi G a^2 \sum_i(\bar\rho_i + \bar P_i)\sigma_i$. The 00 equation,

$$C_{00} \equiv k^2\phi + 3\mathcal{H}(\dot\phi + \mathcal{H}\psi) + 4\pi G a^2 \delta\rho = 0, \tag{2}$$

is not used for evolution; we report it normalised by the sum of the magnitudes of its three terms. CDM, baryons, photons and neutrinos obey the MB95 equations (eqs. 30, 50, 64–65 of the arXiv version, with the baryon equation of their §5.6). Before recombination we use the zeroth-order tight-coupling approximation (TCA): $\theta_b = \theta_\gamma \equiv \theta_{\gamma b}$, $\sigma_\gamma = 0$, and

$$\dot\theta_{\gamma b} = \frac{-\mathcal{H}R\,\theta_{\gamma b} + \frac{1}{4}k^2\delta_\gamma + Rc_s^2k^2\delta_b}{1+R} + k^2\psi, \tag{3}$$

with $R = 3\bar\rho_b/4\bar\rho_\gamma$. After the TCA we evolve the full photon hierarchy with polarisation and Thomson coupling. Deep in the matter era we optionally use the radiation-streaming approximation (RSA) of CLASS [3], $\delta_{\gamma,\nu} = -4\phi$, $\theta_{\gamma,\nu} = 6\dot\phi$, $\sigma = 0$, switched on per mode at $\tau_R(k) = \max(45/k, \tau_{\rm fs})$ exactly as in CLASS.

The background (a, $\mathcal{H}$, $\bar\rho_i$), the Thomson opacity $\dot\kappa = a n_e \sigma_T$, the baryon sound speed, the sound horizon r_s and the Silk scale k_D are taken from CLASS and tabulated as cubic splines that JAX [19] can differentiate. For the parameter bundle of Sec. IV C we use an analytic radiation-plus-matter background that is differentiable in (ω_b, ω_c); it agrees with CLASS to 10^{-9} in a, $\mathcal{H}$ and $\bar\rho_i$ before recombination. The fiducial cosmology is Planck 2018 ΛCDM [20] with massless neutrinos ($N_{\rm eff} = 3.046$) and no reionisation.

A single module implements the right-hand side $f(\tau, u; k)$ of all equations. Both the PINN residual and the reference solver call it, so any disagreement between them is a training error, not a modelling difference.

B. Constraint-consistent initial conditions

MB95 give the adiabatic growing mode in Newtonian gauge to leading order in $k\tau$ and $\tau/\tau_{\rm eq}$. These initial conditions violate Eq. (2) at $\mathcal{O}(\tau_i/\tau_{\rm eq})$. Evolving ϕ with Eq. (1) admits one extra solution beyond those of the full Einstein equations, along which $C_{00} \propto 1/a$; relative to the physical terms it grows like τ in the radiation era. A violation of 2×10^{-5} at $\tau_i = 0.01\,$Mpc grew to 5% by recombination and produced 2–6% errors in every variable compared with CLASS. We therefore use the CLASS initial conditions: the synchronous-gauge expansion to first order in $(k\tau)^2$ and $\omega\tau$ ($\omega = a\bar\rho_m/\sqrt{\bar\rho_r}$), followed by the gauge transformation with $\alpha = (\dot h + 6\dot\eta)/2k^2$ taken from the 00 and $0i$ equations. The initial state then satisfies Eq. (2) to $< 10^{-8}$, and reduces to MB95 eq. 98 to 2×10^{-3}. We verified the propagation property directly: on the constraint surface $dC_{00}/d\tau$ vanishes to 10^{-9} relative for both state layouts.

C. Reference solver and validation against class

The reference solver integrates the same right-hand side with `scipy`'s Radau method ($r_{\rm tol} = 10^{-10}$ unless stated), using exact Jacobians from JAX, with the same TCA/full/RSA switches as the network. Against CLASS (Newtonian gauge, matched $\ell_{\max}$, late-time approximations off) it agrees to $\le 1.1\times10^{-3}$ on ϕ, δ_c, δ_γ for $k = 0.01$ and $0.05\,{\rm Mpc}^{-1}$ up to $2\tau_{\rm rec}$. Without the TCA, every variable agrees to $\sim 2\times10^{-4}$; the remaining difference comes from our zeroth-order TCA and vanishes linearly as the TCA switch is moved earlier.

Our line-of-sight code uses the CLASS Newtonian-gauge source functions and radial functions [21]. Fed CLASS's own source functions it reproduces CLASS C_ℓ^{TT} to $\le 2\times10^{-4}$ (except 1.4×10^{-3} at one multipole). Fed reference-solver sources with the CLASS approximation schedule it reproduces CLASS to $\le 7.7\times10^{-3}$ for $\ell \le 500$. CLASS must be run with a high $\ell_{\max}$ for this comparison, because it ties its k range to $\ell_{\max}$.

III. PINN FORMULATION

A. Windows, hard initial conditions and output scaling

A window network N covers $\tau \in [\tau_a, \tau_b]$ with input $x = \ln(\tau/\tau_a)/\ln(\tau_b/\tau_a) \in [0,1]$, the normalised $\ln k$, and optionally normalised cosmological parameters $\mathbf{c}$. It represents the state of one equation layout (TCA, full hierarchy or RSA) as

$$u(x) = u_a + \left(1 - e^{-x/\epsilon}\right) S(\tau, k) \odot B(x, k, \mathbf{c}), \tag{4}$$

which reproduces the window's initial state u_a exactly [11]. S is the physical magnitude of each variable. It is time-dependent: outside the horizon $\theta \propto k^2\tau$ and $F_\ell \propto (k\tau)^\ell$, rolling over at $k\tau \sim 1$; photon multipoles with $\ell \ge 2$ are additionally suppressed by $(k+\mathcal{H})/(k+\mathcal{H}+\dot\kappa)$ during tight coupling; matter densities grow as $(k\tau)^2$ in

the matter era. Without this scaling the loss is dominated by a $1/(k\tau)^2$ amplification in Eq. (1): the initial loss was 3×10^{11} and training failed. With it the initial loss is $\mathcal{O}(1)$.

B is either the network output or the oscillation ansatz

$$B = N_0 + e^{-k^2/k_D^2}\,(N_1 \cos kr_s + N_2 \sin kr_s) + N_3 \cos k\tau + N_4 \sin k\tau, \tag{5}$$

with smooth network heads N_i. The acoustic phase kr_s and damping envelope carry the photon–baryon oscillations, and $k\tau$ the free-streaming phase. A related basis was used as network *input* features by CosmicNet [4]; here it enters the output structure.

B. Loss

The loss of a window is

$$\mathcal{L} = \Big\langle \frac{1}{n} \sum_i \Big[\frac{\partial_x u_i / L - \tau f_i(\tau, u)}{S_i\, d_i} \Big]^2 + w_c\, C_{00}^2 \Big\rangle, \tag{6}$$

averaged over collocation points, with $L = \ln(\tau_b/\tau_a)$. The first term is the log-time residual of each equation in units of the variable's size. The factor $d_i = 1 + \tau |\partial f_i/\partial u_i|_{\rm Thomson}$ preconditions the stiff Thomson rows (θ_γ, θ_b, $F_{\gamma\ell}$, $G_{\gamma\ell}$), so that quasi-static balances are learned at $\mathcal{O}(1)$ weight. The second term is the normalised 00 residual with $w_c = 1$. It is not optional: it is the only term that penalises the constraint-violating solution of Sec. II B. In RSA windows, and for the modified-gravity model of Sec. IV F, the 00 equation is not an exact consequence of the evolution equations, and we set $w_c = 0$.

C. Time-marching and per-mode switches

A single network over the whole time range fails on oscillating modes (Sec. IV A). We instead train a chain of windows in sequence. Window j starts from the frozen window $j-1$ evaluated at its end, mapped between layouts where they differ. At the TCA→full switch we hand over the first-order tight-coupling state: the baryon–photon slip from the quasi-static baryon Euler equation, $\sigma_\gamma = \frac{16}{45}\theta_\gamma/\dot\kappa$ and the corresponding polarisation moments. Handing over zero slip starts a transient of width $1/\dot\kappa$ that the next window cannot represent; its loss was three orders of magnitude above that of every other window. The TCA→full switch is at a fixed τ at which the TCA holds for all modes in the bundle. The full→RSA switch depends on k, and we found it must: switching all modes to the RSA at a common time ruins the low-ℓ spectrum (Sec. IV E). Each mode is therefore trained in the full-hierarchy windows only up to its $\tau_R(k)$, and the RSA windows are parameterised in $\ln(\tau/\tau_R(k))$, starting from the full chain evaluated at $\tau_R(k)$.

D. Training

Each window is an MLP (4 hidden layers of 64–96 tanh units) trained with Adam (learning rate 10^{-3}, cosine decay) on resampled collocation sets, followed by L-BFGS on a fixed, denser set. Collocation is a tensor product of stratified $\ln k$ (and Latin-hypercube parameters) with stratified window times. Training is in float64 on CPU. No reference solution enters training.

IV. RESULTS

A. Controlled experiments: what makes training work

We first trained single modes in the tight-coupling era, comparing each variant with the reference solver integrating the identical equations. Errors are $\max_\tau |X_{\rm PINN} - X_{\rm ref}| / \max_\tau |X_{\rm ref}|$ for each variable X; this metric is well defined through zero crossings, unlike pointwise relative error.

a. Output scaling. With fixed per-variable scales the initial loss was 3×10^{11} and training made no progress. The $1/(k\tau)^2$ factor in Eq. (1) amplifies any error in θ outside the horizon. Scaling the outputs by their super-horizon growth reduced the initial loss to $\mathcal{O}(1)$.

b. The constraint term. At $k = 0.01\,\mathrm{Mpc}^{-1}$, up to $\tau = 200\,\mathrm{Mpc}$, training on the evolution residuals alone reached a lower ODE loss (3.8×10^{-6}) than training with the 00 term, yet its errors were 30 times larger: 2.7×10^{-2} on ϕ against 9.7×10^{-4} (Fig. 1). Its error grows steadily after $\tau \sim 10\,\mathrm{Mpc}$, the signature of the constraint-violating solution of Sec. II B. With the constraint term and a longer L-BFGS polish (12k steps), every variable reached $\leq 6 \times 10^{-4}$ (ϕ: 8.5×10^{-5}) in 9 CPU-minutes. Causal weighting [22] at $\epsilon = 1$ had no effect: the weights stayed ≈ 1.

c. Oscillations and time-marching. At $k = 0.2\,\mathrm{Mpc}^{-1}$, with the tight-coupling equations integrated to $\tau = 200\,\mathrm{Mpc}$ (3.4 acoustic oscillations; a numerical stress test, since the TCA is not physically valid there), a single network failed with every error $\approx 100\%$. It relaxed all variables towards zero from τ_i onwards (Fig. 2, left), with or without the oscillation ansatz. Causal weighting at $\epsilon = 10$ and 100 did not help. This is not a problem of the equations or of capacity: the super-horizon segment alone ($\tau \leq 1\,\mathrm{Mpc}$) trained to $\sim 2 \times 10^{-3}$ at the same k. The single network gives up accuracy at early times to reduce the larger late-time oscillatory residual, a failure mode known for PINNs [23]. Time-marching, with one window up to $k\tau = 1$ and then one per acoustic period, chained by hard initial conditions [23, 24], fixed it: every variable reached $\leq 1.0 \times 10^{-3}$, with no visible accumulation of error across the 8 windows (Fig. 2, right). At $k = 0.05$, where the TCA era holds half an oscillation, the oscillation ansatz improved a single network by $\approx 2\times$.

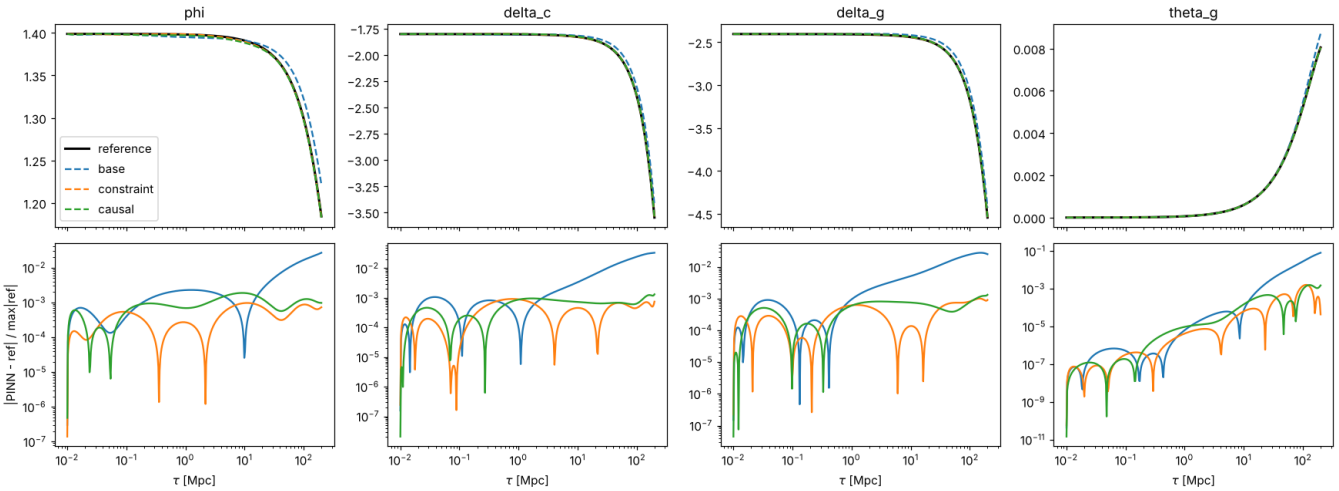


FIG. 1. Single mode $k = 0.01\,\mathrm{Mpc}^{-1}$, tight-coupling era. Top: PINN variants against the reference solver. Bottom: error normalised by $\max|X_{\mathrm{ref}}|$. Without the 00 loss term (blue) the error grows steadily after $\tau \sim 10\,\mathrm{Mpc}$, despite a lower ODE residual.

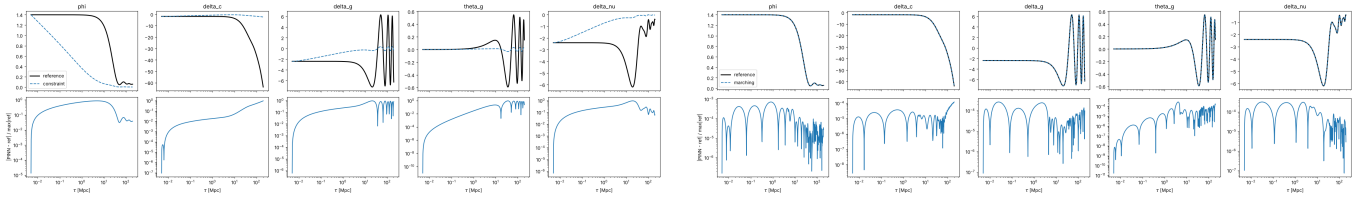


FIG. 2. $k = 0.2\,\mathrm{Mpc}^{-1}$ stress test through 3.4 acoustic oscillations. Left: a single network (with oscillation ansatz) collapses towards zero from the first time step. Right: time-marching over 8 windows; every variable is within 10^{-3} of the reference.

d. The tight-coupling switch. In a single-mode chain through recombination, the first full-hierarchy window trained to a loss of 6.7×10^{-2}, against $\sim 5 \times 10^{-5}$ for every later window, when the state was handed over with zero baryon–photon slip. The full system relaxes to the quasi-static slip over $\delta\tau \sim 1/\dot{\kappa} \approx 0.01\,\mathrm{Mpc}$, a boundary layer the network cannot represent. Handing over the first-order slip places the state on the slow manifold; the handed-over slip then matches the full system's to 10^{-4}.

B. Phase 1: wavenumber bundle in the tight-coupling era

A three-window chain (edges 10^{-3}, 1, 20, 200 Mpc; 4×64 networks with the oscillation ansatz; 20k Adam + 6k L-BFGS steps per window) was trained over $k \in [10^{-3}, 0.05]\,\mathrm{Mpc}^{-1}$. It was evaluated on 20 held-out k, each up to its own TCA switch time. The RMS error over modes is $(0.9\text{–}2.2) \times 10^{-3}$ across the variables, with maxima of $(1.7\text{–}5.1) \times 10^{-3}$. The worst variable is the neutrino shear, and the worst mode is $k = 0.033$. This misses the 10^{-3} target by a factor of 2–5. The error is already $\sim 10^{-3}$ in the first, super-horizon window (Fig. 3), where the solution hardly depends on k. It therefore reflects accumulated residual error over a long window, not missing capacity for the k dependence. Shorter windows, wider networks and second-order optimisers are the obvious remedies [**Pending:** retrained Phase 1 chain].

C. Phase 2: cosmological-parameter bundle

A six-window chain (edges 10^{-3}, 1, 5, 20, 50, 100, 200 Mpc; 4×96 networks with the oscillation ansatz and (ω_b, ω_c) as extra inputs; 20k Adam + 6k L-BFGS steps per window) was trained over $(k, \omega_b, \omega_c) \in [10^{-3}, 0.2]\,\mathrm{Mpc}^{-1} \times [0.020, 0.024] \times [0.10, 0.14]$ on the analytic background. Each mode is trained to 1.25 times its fiducial TCA switch time. Final window losses rise from 6×10^{-7} in the first window to 4×10^{-5} in the last. The chain was evaluated on a held-out Latin hypercube of 30 points, each up to its own switch time, against the reference solver on the same background. The RMS error over points is $(5\text{–}13) \times 10^{-3}$ across the variables, and the maxima are $(1.4\text{–}6.3) \times 10^{-2}$. The largest error is in δ_b at $k = 0.2\,\mathrm{Mpc}^{-1}$ and $\omega_c = 0.136$. Phase 2 therefore misses its 10^{-3} target by an order of magnitude, and does worse than the k-only bundle of Phase 1.

The errors are largest at the edges of the parameter

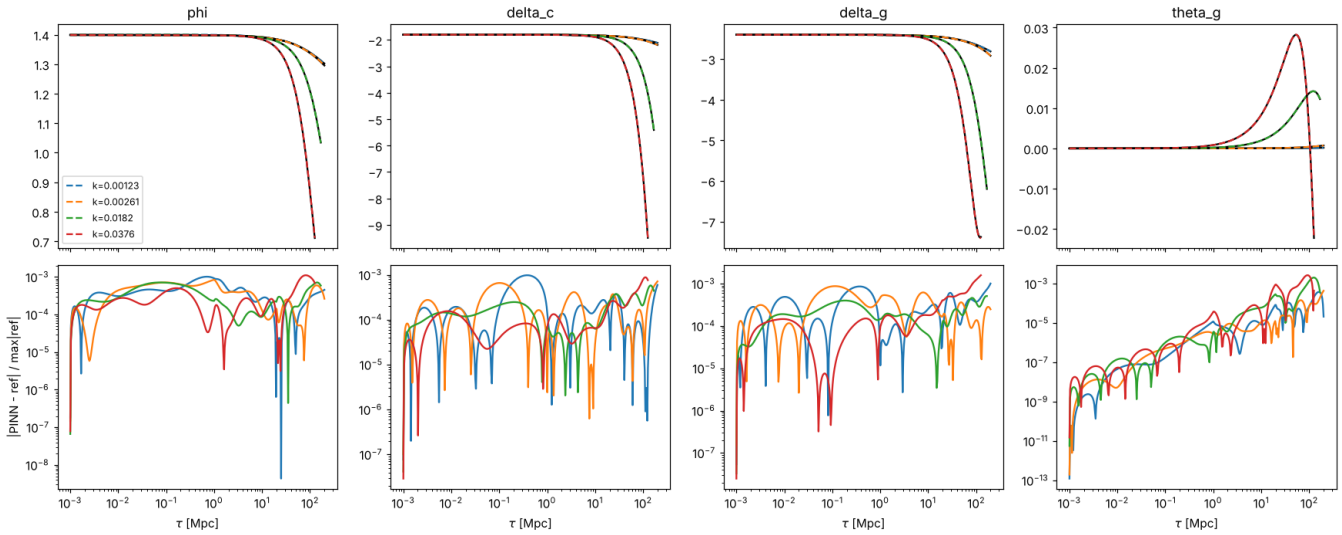


FIG. 3. Phase 1 bundle over $k \in [10^{-3}, 0.05]\,\mathrm{Mpc}^{-1}$: four of the 20 held-out modes. Bottom: error normalised by $\max|X_{\mathrm{ref}}|$.

box (Fig. 4, left). At $k = 1.1 \times 10^{-3}$, the error in ϕ is 4×10^{-3} already in the first, super-horizon window at $\omega_c = 0.138$, but only 1×10^{-3} at $(\omega_b, \omega_c) = (0.022, 0.12)$. The error therefore comes from uneven fit quality across the conditioning inputs, not from error accumulating along the chain. The bundle loss is an average over the box, so the corners are fitted worst. The constraint residual tracks the error: over the 30 test points, the correlation between $\log \max_\tau |C_{00}|$ and the log of the maximum error is 0.74 (Fig. 4, right). The worst point, at $k = 0.2\,\mathrm{Mpc}^{-1}$, also has the largest residual ($|C_{00}| = 0.11$). C_{00} is never used as a training target at test points, so it is a usable, label-free indicator of where a bundle cannot be trusted. Training more points near the box edges, or weighting the loss by the local residual, are the natural next steps.

D. Phase 3: through recombination

[**Pending:** 25-window chain (3 TCA, 19 full-hierarchy with $\ell_{\max} = 12/10/17$, 3 per-mode RSA) over $k \in [5 \times 10^{-6}, 0.12]\,\mathrm{Mpc}^{-1}$; errors against CLASS up to $2\tau_{\mathrm{rec}}$.]

E. Phase 4: C_ℓ^{TT}

The approximation schedule matters before the network does. With reference-solver sources, the CLASS-like schedule (TCA trigger; RSA per mode at $k\tau > 45$) reproduces CLASS C_ℓ^{TT} to $\leq 7.7 \times 10^{-3}$ for $\ell \leq 30$ and $\leq 2 \times 10^{-3}$ above. A k-independent RSA switch, which would let a bundle use fixed windows, fails at low ℓ. Switching all modes at $\tau = 850\,\mathrm{Mpc}$ gives C_2 wrong by a factor of 200, and at $\tau = 1400\,\mathrm{Mpc}$ 13% errors for $\ell \leq 5$; both are fine ($\leq 1\%$) above $\ell \sim 30$. Low-k modes never satisfy $k\tau > 45$ and need the full hierarchy until today, which is why the PINN chain switches per mode. [**Pending:** C_ℓ^{TT} from PINN source functions vs CLASS, $\ell \leq 500$.]

F. Phase 5: a late-time modification of gravity and an inverse problem

We consider $\mu(a) = G_{\mathrm{eff}}/G = 1 + \mu_0\,\Omega_\Lambda(a)/\Omega_{\Lambda,0}$ multiplying G in the perturbed Einstein equations, with an unchanged background. This phenomenological model is not available in standard CLASS. Since $|\mu - 1| < 2 \times 10^{-5}|\mu_0|$ before $\tau = 2400\,\mathrm{Mpc}$, the Phase 3/4 chain is reused up to there, and only the later windows are retrained with $\mu_0 \in [-0.5, 0.5]$ as an extra input. In the reference solver, $\mu_0 = 0.3$ raises the comoving matter density today by $\approx 1\%$ and the potential by $\approx 10\%$, independently of scale. [**Pending:** validation of the μ_0 bundle; recovery of μ_0 from mock $z = 0$ matter and Weyl-potential spectra (1% errors), PINN vs reference-solver likelihoods and their cost.]

V. DISCUSSION

a. What generalises. Several of the ingredients above are not specific to PINNs. The growth of the constraint-violating solution under $0i$ evolution affects any solver that evolves ϕ with Eq. (1). It is harmless only when the initial conditions satisfy the 00 constraint to the order of the neglected terms. For PINNs specifically, a small residual does not imply a small error in this system: the 00 residual must be part of the loss, and it serves as a free *a posteriori* diagnostic. Every failure we met (trivial solutions, the boundary layer at the TCA switch, constraint drift) is visible in the per-window

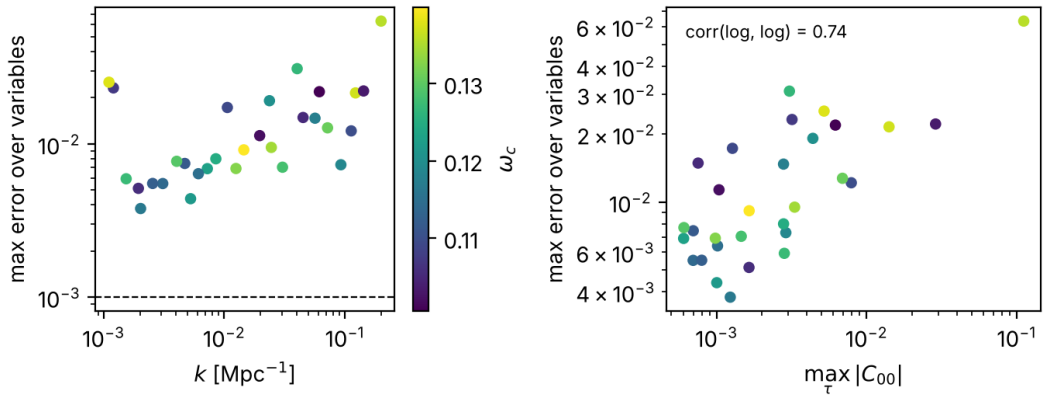


FIG. 4. Phase 2 bundle over (k, ω_b, ω_c): maximum relative error over variables at 30 held-out points (left), and the same error against the maximum Einstein-00 residual along each trajectory (right). Colour: ω_c. Dashed: the 10^{-3} target.

losses or in C_{00} without reference to the true solution.

b. Cost. On a 4-core CPU in float64, one tight-coupling window of the k bundle trains in 20–40 minutes, and the full chain to today in [**Pending:** hours]. The reference solver integrates one mode in 5–10 s and CLASS computes C_ℓ^{TT} in seconds. As a forward solver for a single cosmology, the PINN is therefore not competitive with existing codes, and we do not claim otherwise. Once trained, a bundle evaluates the late-time state of a mode in milliseconds, with exact gradients with respect to k and conditioning parameters. Its value lies in amortisation over many evaluations of models for which no training data exist (Sec. IV F). We have not compared against differentiable solvers [8–10] on equal hardware [**Pending:** benchmark].

c. Limitations. The present results use zeroth-order tight coupling, massless neutrinos and no reionisation, and stop at the unlensed temperature spectrum. The Phase 1 bundle misses its 10^{-3} target by a factor of 2–5, and the Phase 2 parameter bundle by an order of magnitude. All runs use one random seed per configuration. Residual-based error bounds [16, 25, 26] would turn the C_{00} diagnostic into a certificate; extending them to this linear system is left for future work.

d. Conclusions. [**Pending:** to be written with the final results.]

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